MODUCARD SYSTEM 3

HANDBOOK

MODULAR MECHANICA 2026

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MCS 3

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Contents

1 Introduction

1.1 About This Handbook

This handbook describes the ModuCard System 3 (**MCS 3**) interface standard — a system of interlocking modular robotic cards that can be freely combined, extended, and reconfigured without tools.

Note: Read this chapter in full before assembling any modules. Incorrect insertion order may damage bus termination circuits.

The document is structured in four self-contained parts. Each part can be read independently; cross-references indicate dependencies where they exist.

1.2 What Is ModuCard System 3?

MCS 3 is a family of interchangeable robotic building blocks. Every module exposes four standardised connector faces (N, S, E, W) and communicates over a shared differential bus. Modules are hot-swappable: the bus protocol detects insertion and negotiates addresses automatically.

Core properties at a glance:

- **Power:** 24 V shared rail, 3 A per face maximum
- **Data:** 10 Mbit/s half-duplex serial, CRC-16 framing
- **Addressing:** automatic negotiation on hot-attach
- **Topology:** tree or daisy-chain, up to 63 modules

1.3 Conventions Used in This Handbook

`Monospace`
Register names, bus commands, and literal byte values.

Bold sans
First use of a defined term; see Glossary (Appendix A).

[ranges]
Square brackets denote inclusive value ranges, e.g. `[0x00–0xFF]`.

Dimensions
All dimensions in millimetres at 20 °C unless otherwise stated.

Important: Chapters marked ⋆ in the Table of Contents contain normative requirements (shall / shall not). All other chapters are informative.

1.4 How To Use This Handbook

If you are a **hardware designer**, start with Part I (Mechanical) and Part II (Electrical), then refer to Part IV (Conformance) for test procedures.

If you are a **firmware developer**, start with Part III (Communication Protocol), where the complete packet format and API are defined.

If you are a **system integrator**, Part I Chapter 3 (Assembly Rules) and Part III Chapter 9 (Topology Planning) are the most immediately useful sections.